

ASIA south

Himalayas



LOCATION Running southeast from northern Pakistan and India across Nepal to Bhutan

TYPE Continent–continent collision

LENGTH 2,400 miles (3,800 km)

The Himalayas are the highest mountain range on Earth, as well as one of the youngest. If the neighboring Karakoram Range is included, the Himalayas contain Earth's 14 highest mountain peaks, each with an altitude of over 5 miles (8 km), including its highest mountain, Mount Everest. These peaks are still being uplifted at the rate of some 20 in (50 cm) per century by the continent–continent collision that originally formed them. However, the mountains are weathered and eroded at almost the same rate, with the debris carried away by great rivers, such as the Ganges and Indus to the south.

The collision that brought about both the Himalayas and the Tibetan Plateau to its north occurred between 50 and 30 million years ago, when tectonic plate movements caused India—at that time an island continent—to crash into Southeast Asia. For millions of years before the collision, the floor of the ocean between India and Asia (called Tethys) was consumed by subduction under the Eurasian Plate. But once the ocean closed, first the continental margins between India and Asia, and finally the continents themselves, collided. The crust from both was thickened, deformed, and metamorphosed and parts of both continents and the floor of the Tethys Ocean were pushed up to form the



Himalayas. Today, because the Himalayas are still rising, earthquakes and accompanying landslides remain a common occurrence.

The mountains form a number of distinct ranges. Traveling northward,

from the high plains of the Ganges, the first of these are the Siwalik Hills, a line of gravel deposits carried down from the high mountains. Here, there are subtropical forests of bamboo and other vegetation. Farther north are the Lesser Himalayas, which rise to heights of about 3,000 ft (5,000 m) and are traversed by numerous deep gorges formed by swift-flowing streams.

Farthest north are the

Great Himalayas, between 20,000 and 29,000 ft (6,000 and 8,800 m) tall and containing the highest peaks.

This region is heavily glaciated and contains lakes filled with glacial meltwater.



EASTERN HIMALAYAS

In this satellite view of an eastern region of the Himalayas, which extends into China, the snow-covered high-altitude regions are clearly delineated.

MOUNT EVEREST

At 29,035 ft (8,850 m), Everest is the highest peak on Earth. Satellite studies show it is being uplifted by a fraction of an inch per year.



TIBETAN RANGE

The Kailas Range is a central region of the Himalayas, close to the border between Tibet and India. Here, the mountains are viewed from the Tibetan Plateau, which is itself about 3 miles (5 km) above sea level.

GLACIAL LAKES

Many of the higher areas of the Himalayas are covered in glaciers and dotted with lakes dammed by glacial moraines. In the left foreground is the Tsho Rolpa Glacier Lake in northeast Nepal, which, at 15,092 ft (4,600 m), is one of the highest lakes on Earth.